Q-TRACE · November 10, 2026 · Policy selection

One-page summary · meeting 5 of 7 · Phases 5A–5C, completed October 7–8, 2026

Chronology. Completed on Oct 7–8, 2026; the sealed test followed on Oct 8 and is the Nov 24 topic.

Purpose. To explain why a learned controller was originally proposed, how the 64-model benchmark and label-free probes were built, and why a simple rule emerged instead.

Key numbers.

Item	Result	Source
Model pool	64 VQCs = 16 structures × 4 seeds; median AUPRC 0.235 (0.047–0.538); split 8/4/4	`p5a_pool_metrics.csv`, `p5b_split.csv`
Probes	66 circuits per batch; readout probe $\rho = 0.999$; gate probe $\rho = -0.89$ (depolarizing), -0.12 (thermal)	`p5c_probe_validity.csv`
Phase 5B (categorical)	4-cell rule wrong in 0.33% of validation cases (best single action wrong in 62.5%)	`p5b_metadata.json`
Phase 5C (continuous)	combined mitigation 67–71%, ZNE 23–27% of contexts; budget changes the choice < 3%; 3× shots never chosen	`p5c_metadata.json`, `p5c_budget_effect.csv`
Best rule (validation)	9.6% disagreement, 2.0% [1.6, 2.7] material, Brier regret +0.3%	`p5c_policy_eval.csv`
Gates	G3B fail, G3C fail; residual detectable (MI 0.08 / 0.04 nats) but not material	`p5b_metadata.json`, `p5c_metadata.json`

The rule.

- If the gate probe detects gate noise, use ZNE, and add readout mitigation when the readout probe exceeds about 0.4%.
- Otherwise use readout mitigation alone. At 256 shots, use no mitigation if the readout probe is also below its threshold.

Takeaways.

- As registered, no learned controller was trained.
- In our design, 16 of 17 gate settings carry gate noise, so the material decision is whether gate noise is present.

Slide	Content	Speaker (proposed)
1	Title and message	Mayank
2	The decision we proposed to learn	Mayank
3	64 models and two probes	Mayank
4	Categorical, then continuous noise	Tushar
5	Gates failed; a simple rule emerged	Tushar

Needs confirmation.

- The presenter split.
- The meeting date and format: supplied by the team, not on a public course calendar.